Homework One DUE 10/11

(1) $P(A \cup B \cup C)$ can be broken down by associative property

Let $(A \cup B) = X$

$P(A \cup B \cup C) = P(X \cup C) = P(X) + P(C) - P(X \cap C)$

- Replace X
- given in class
- distribute

$= P(A \cup B) + P(C) - P((A \cup B) \cap C)$

$= P(A) + P(B) - P(A \cap B) + P(C) - P((A \cup B) \cap C)$

$= P(A) + P(B) - P(A \cap B) + P(C) - P((A \cap C) \cup (B \cap C))$

$= P(A) + P(B) + P(C) - P(A \cap B) - [P(A \cap C) + P(B \cap C) - P((A \cap C) \cap (B \cap C))]$

$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

□ ✓

(2) 1) $\underline{26}\ \underline{26}\ \underline{10}\ \underline{10}\ \underline{10}\ \underline{10}$ $26^2 \cdot 10^4$ 2) $\underbrace{\quad}_{26}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}$ $26^3 \cdot 10^3$

(3) 1) $\dfrac{\binom{13}{5}\binom{13}{4}\binom{13}{3}\binom{13}{1}}{\binom{52}{13}}$ 2) $\dfrac{\binom{13}{5}\binom{13}{4}\binom{13}{2}\binom{13}{2}}{\binom{52}{13}}$

3) $\dfrac{\binom{13}{5}\binom{13}{4}\binom{13}{1}\binom{13}{3}}{\binom{52}{13}}$

(4) 1) $\frac{13}{52} \cdot \frac{12}{51}$ 2) $\frac{13}{52} \cdot \frac{13}{51}$

3) $A = \heartsuit$ on first draw $B = A$ on first draw $C = A$ on second draw

$P(C \cap (A \cap B)) + P(C \cap (A \cap B')) =$

$P(C \mid A \cap B) \cdot P(A \cap B) + P(C \mid A \cap B') \cdot P(A \cap B') =$

$\left(\frac{3}{51}\right)\left(\frac{1}{52}\right) + \left(\frac{4}{51}\right)\left(\frac{12}{52}\right) = \frac{3+48}{2652} = \frac{51}{2652} = \frac{1}{52}$

C order not matter

(5) 1) 365^r 2) ${}_{365}P_r = \frac{365!}{(365-r)!}$

3) A = ≥2 students have same bday

A' = everyone has unique bday

$P(A) = 1 - P(A')$

$P(A) = 1 - \frac{{}_{365}P_r}{365^r}$

(6) 1) 2 red, 4 yellow draw 5 balls w/o replacement

R4R4R

4R4R4

B = NO 2 balls drawn consecutively are same

A_R = first red A_Y = first yellow

$P(B) = P(B|A_R)P(A_R) + P(B|A_Y)P(A_Y)$

$\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right) =$

$\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3$

2) since we are not replacing, the denominator should be going down too, as we have less total balls each time. Also, only 4R4R4 is possible since we need 3 red balls if we start with red and we only have 2 without replacing.

4 R 4 R 4

$\frac{\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right)\left(\frac{1}{3}\right)\left(\frac{2}{2}\right)}{{}_5P_5}$ $\frac{24}{360}$

accounting for ways to pick (total outcomes) since replacing